Chapter 5:OLAP

What is Dimensional Modeling?

- Dimensional modeling represents data with a cube operation, making more suitable logical data representation with OLAP data management.
- consist of **"fact"** and **"dimension"** tables.
- the transaction record is divided into either **"facts,"** which are frequently numerical transaction data,
  or **"dimensions,"** which gives context to the facts.
- For example, a sale transaction can be damage into **facts** such as the number of products ordered and the price paid for the products,
- **dimensions** such as order date, user name, product number, order ship-to, and bill-to locations, and salesman responsible for receiving the order.

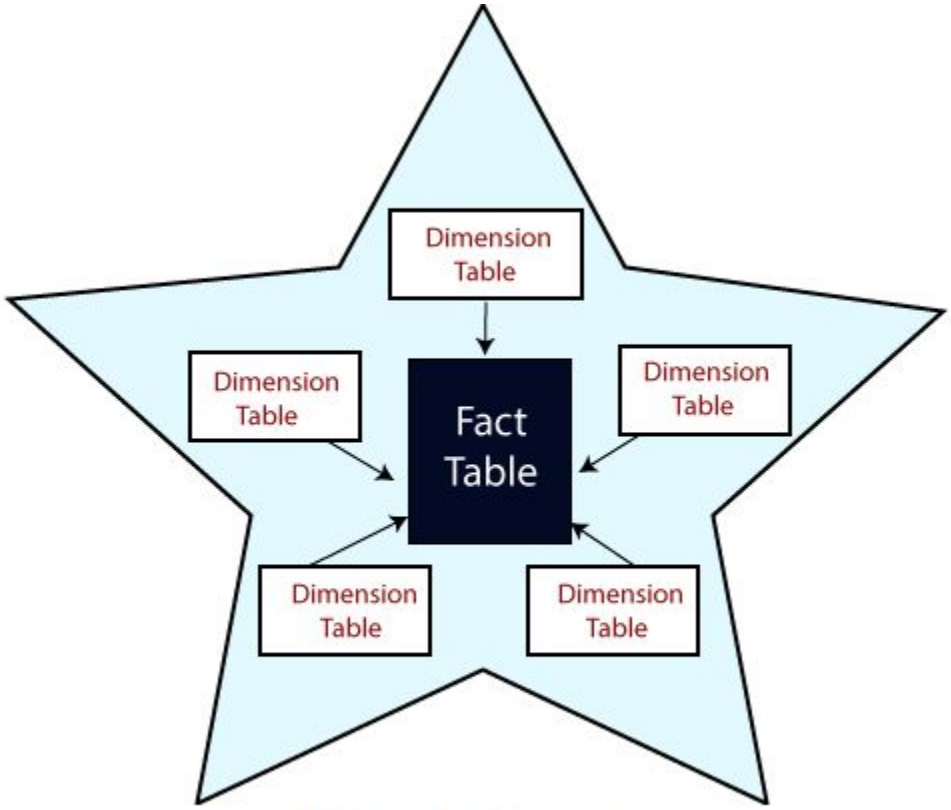
Feature	Fact Table	Dimension Table
Data Type	Quantitative (numerical)	Qualitative (descriptive)
Purpose	Store measurable data for analysis	Provide context and descriptive information
Examples	Sales amounts, transaction counts	Customer names, product categories, dates

Elements of Dimensional Modeling

- **Fact**
- It is a collection of associated data items, consisting of measures and context data. It typically represents business items or business transactions.
- **Dimensions**
- It is a collection of data which describe one business dimension.
- Dimensions decide the contextual background for the facts, and they are the framework over which OLAP is performed.
- **Measure**
- It is a numeric attribute of a fact, representing the performance or behavior to the dimensions.
- two basic models which are used in dimensional modeling:
- Star Model
- Snowflake Model

Star Schema

- Most popular schema design
- Dimensions are stored in dimension table and every entry has its own unique identification
- DT is related to FT
- All the PK of DT become Composite key in FT
- Fact table contains facts
- Eg:Store_id,date_key and product_id giving the amount of certain product sold on given day at given store.
- FK in DT are contained in FT
- Size of FT is large then DT


Dimension Table
Dimension Table
Fact Table
Dimension Table
Dimension Table
Dimension Table

Star Schema

- Star Schema
- Each dimension in a star schema is represented with only one-dimension table.
- This dimension table contains the set of attributes.
- The following diagram shows the sales data of a company with respect to the four dimensions, namely time, item, branch, and location.
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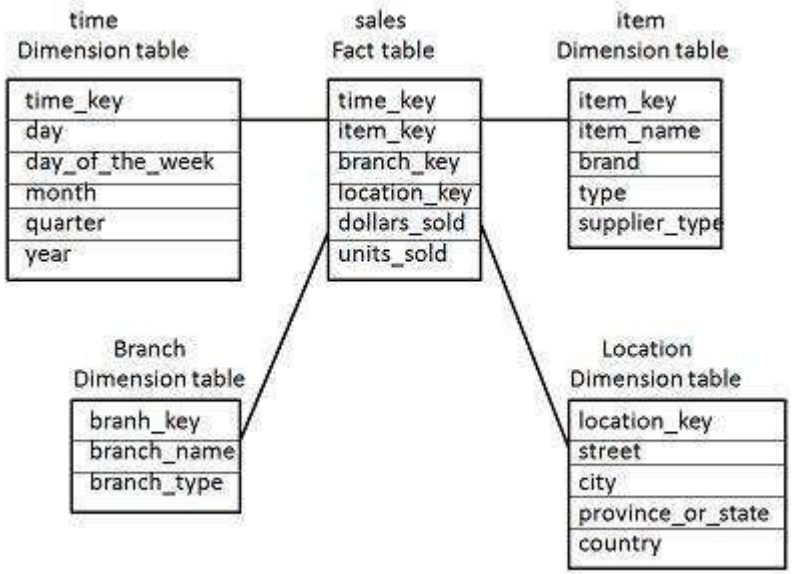
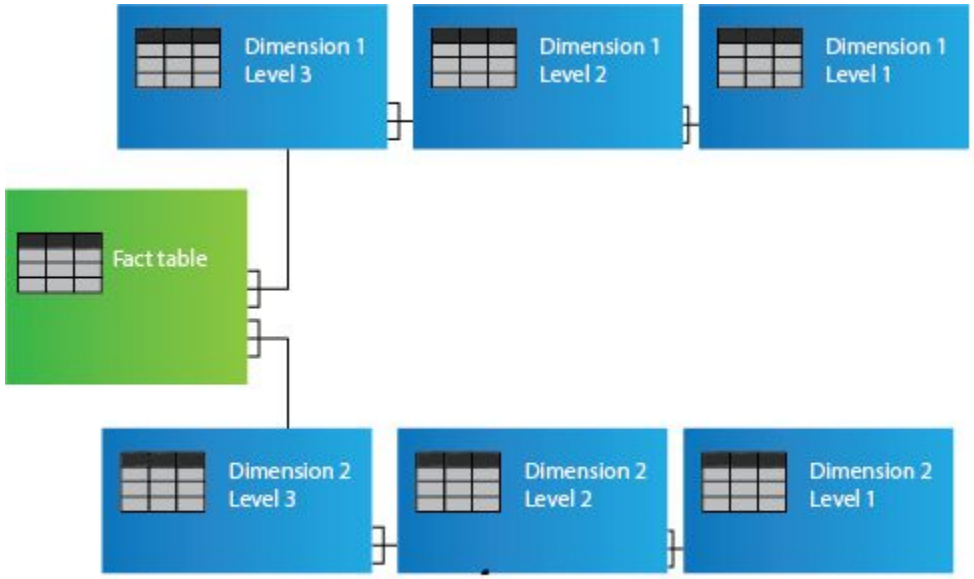


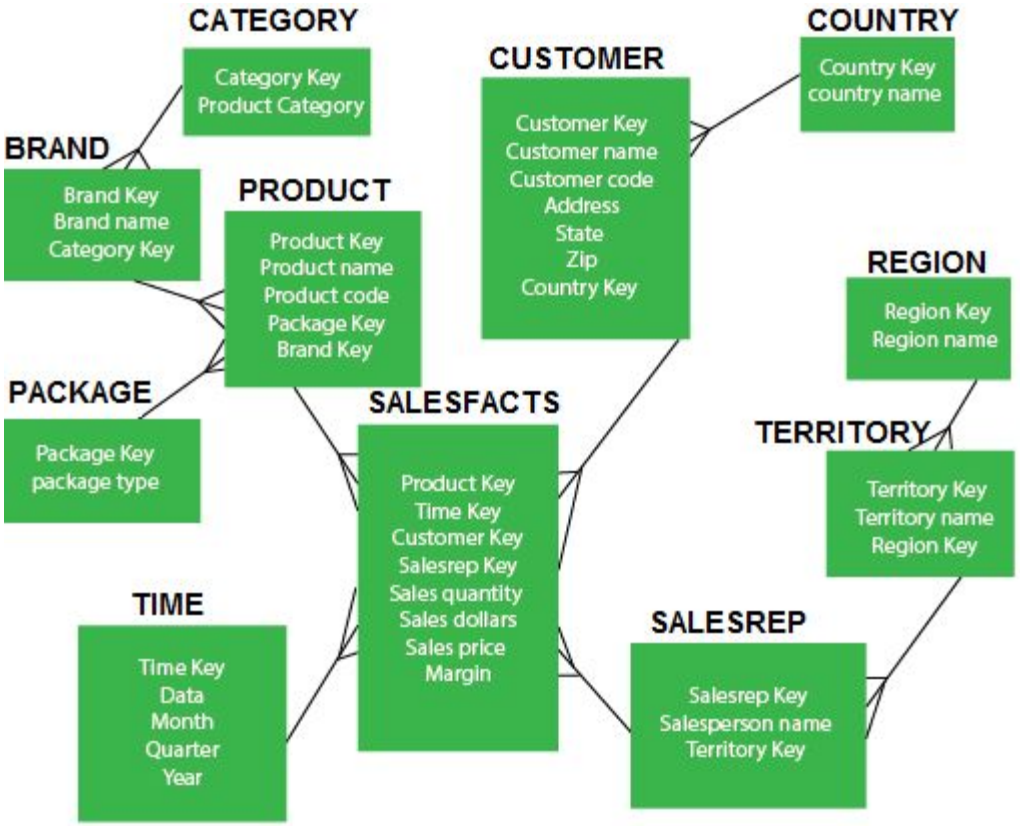
- There is a fact table at the center. It contains the keys to each of four dimensions.
- The fact table also contains the attributes, namely dollars sold and units sold.

Snow Flake Schema

- A schema is known as a snowflake if one or more dimension tables do not connect directly to the fact table but must join through other dimension tables.
- The schema is diagramed with each fact surrounded by its associated dimensions, and those dimensions are related to other dimensions, branching out into a snowflake pattern.
- The snowflake schema consists of one fact table which is linked to many dimension tables, which can be linked to other dimension tables through a many-to-one relationship.

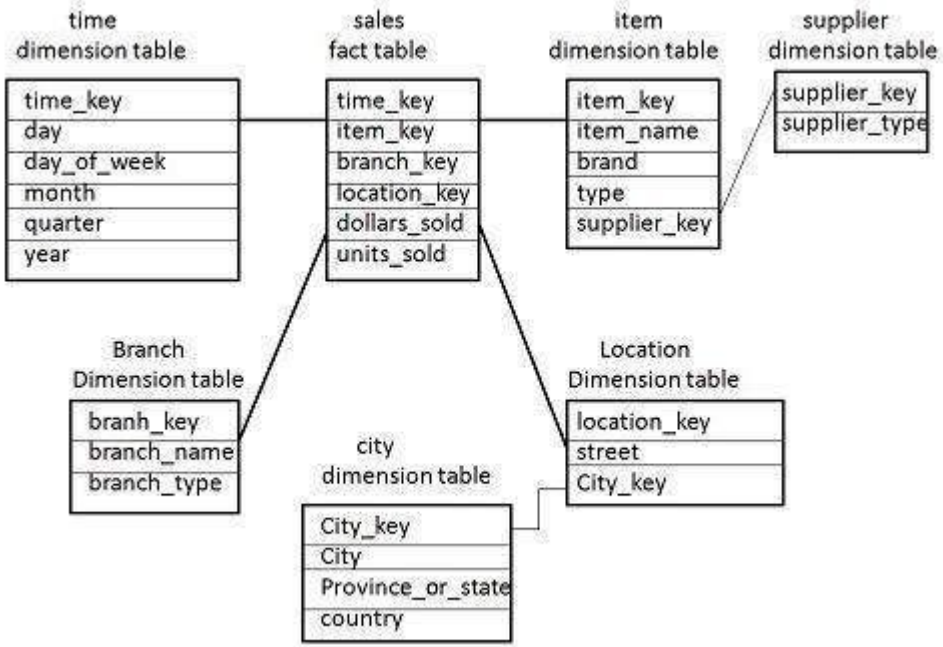


Snowflake Schema

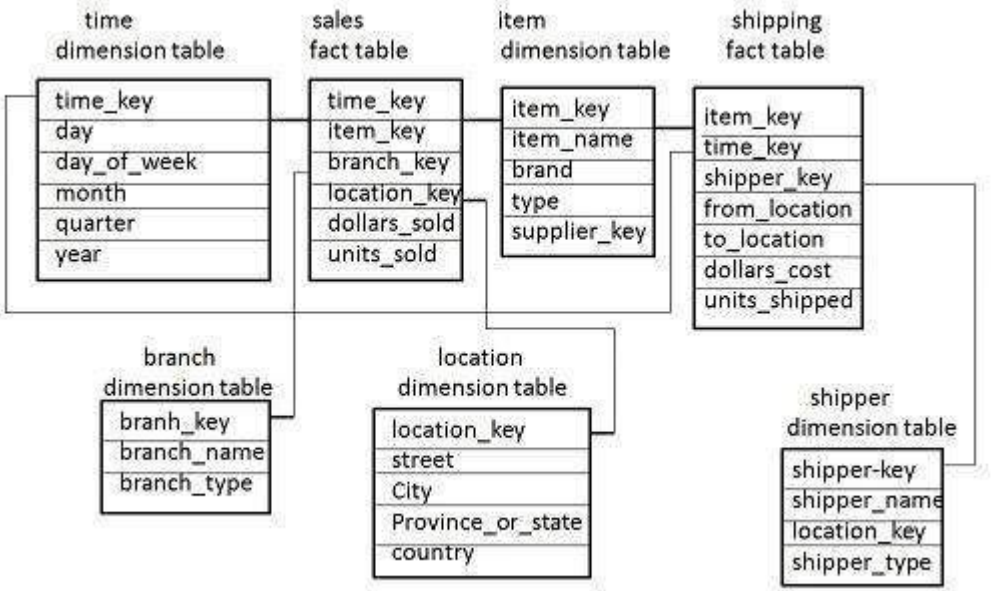


Snowflake Schema

- Snowflake Schema
- Some dimension tables in the Snowflake schema are normalized.
- The normalization splits up the data into additional tables.
- Unlike Star schema, the dimensions table in a snowflake schema are normalized. For example, the item dimension table in star schema is normalized and split into two dimension tables, namely item and supplier table



Fact Constellation Schema

- A fact constellation has multiple fact tables. It is also known as galaxy schema.
- The following diagram shows two fact tables, namely sales and shipping.
- 

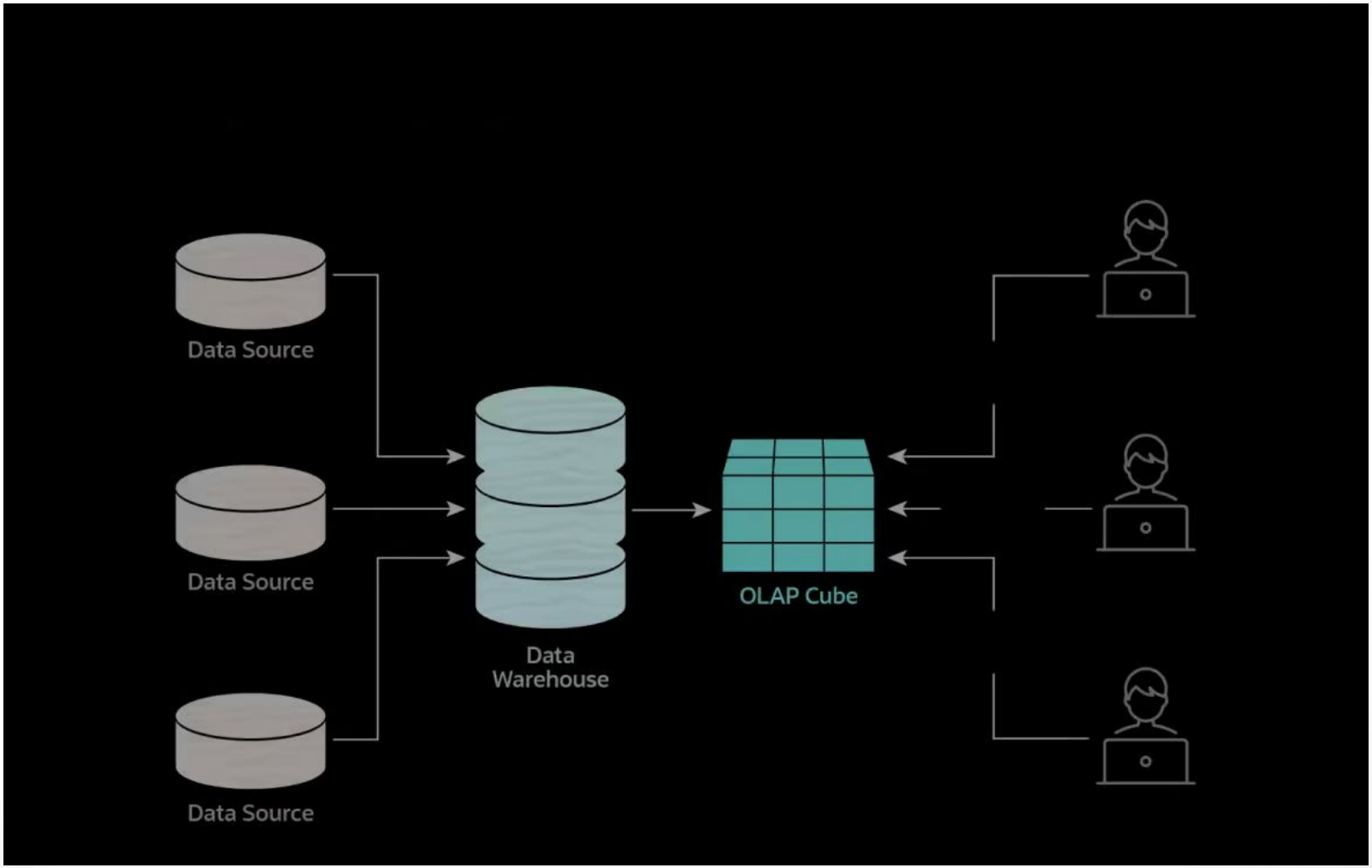
SS vs SF

Star Schema	Snowflake Schema
Top-down model	Bottom-up model
Less time for query execution	More time for query execution
Occupies more space	Uses less space
Denormalized is used	Both normalization and denormalization are used
Low query complexity	Higher query complexity
Less number of foreign keys	More number of foreign keys
High data redundancy	Low data redundancy
Simple design	Complex design
Easy to understand	Hard to understand

What is OLAP

- OLAP (online analytical processing) is a computing method that enables users to easily and selectively extract and query data in order to analyze it from different points of view.

Need for OLAP

- Fast,staedy and proficient access to various views of information
- Complex queries can be processed
- Easy to analyze info by processing complex queries
- OLAP systems are designed to help businesses uncover hidden patterns, trends, and insights within their data, which is crucial for making informed decisions.
- By pre-calculating and integrating data, OLAP systems **reduce the time required for data analysis**, empowering users to focus on insights rather than data preparation
- OLAP systems provide a unified platform for **accessing data from various sources**, ensuring that all business units have a consistent view of the data.

How OLAP Systems Work